

Cross-Cultural Validation and Ideological Fairness of a Historical Perspective Taking Instrument: Evidence from Czech Secondary Students

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Abstract

This study reports ~~the first~~ a Czech adaptation of the Hartmann and Hasselhorn Historical Perspective Taking (HPT) instrument and examines whether its politically charged scenario introduces construct-irrelevant variance (CIV) related to students' ideological attitudes. Among 293 Czech secondary students, confirmatory factor analysis ~~(CFA) confirmed the~~ avored a correlated three-factor ~~structure. Item Response Theory~~ ~~(IRT)-based differential item functioning (DIF)~~ representation. An item-level bias analysis detected no ~~biased items~~ differential item functioning across ideology groups with this sample and procedure. Multi-group ~~CFA demonstrated full scalar measurement~~ invariance factor analysis produced fit-index changes compatible with scalar invariance,

but small extreme groups and unmodelled clustering limit the strength of that evidence. Historical knowledge ~~predicted HPT scores; ideology was unrelated~~ correlated with the primary six-item contextualization composite ($r = .34$, 95% CI [.23, .43]); the combined ideology score was unrelated ($r = -.05$, 95% CI [-.16, .07]). These findings support a ~~transferable~~ possible protocol for evaluating measurement fairness when assessment content carries attitudinal valence, subject to the limits of this single scenario, item format, and ideologically restricted sample.

Keywords: historical perspective taking, cross-cultural validation, measurement invariance, differential item functioning, construct-irrelevant variance, test fairness

1. Introduction

Cross-cultural adaptation of educational and psychological instruments requires more than linguistic translation: it demands evidence that the adapted version measures the same construct, with equivalent psychometric properties, in the target population (American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 2014; Hambleton, Merenda, & Spielberger, 2005; International Test Commission, 2017). When assessment content carries attitudinal valence—when scenarios, stimuli, or items touch on beliefs that respondents may hold—an additional fairness question arises: does the content introduce construct-irrelevant variance (CIV) by functioning differently for subgroups whose attitudes align with that content (Messick, 1995)? This question is especially pressing for scenario-based instruments in the social sciences, where rich contextual material is needed to elicit

40 complex reasoning but may simultaneously activate respondents' pre-existing attitudes.
41 Historical perspective taking (HPT)—the capacity to reconstruct the beliefs, values, and
42 circumstances that made past actors' decisions intelligible in their own time (Hartmann
43 & Hasselhorn, 2008; Lee & Ashby, 2001)—provides a particularly informative case. HPT
44 instruments routinely use politically and morally charged historical scenarios as the
45 stimulus for disciplinary reasoning (reasoning according to the methodological norms of
46 the historical discipline, such as source criticism and ~~contextualisation~~contextualization;
47 van Drie & van Boxtel, 2008; Seixas & Morton, 2013; Wineburg, 2001).

48 The instrument ~~developed by is among the most extensively validated standardized~~
49 ~~HPTmeasures in European research. Originally reported by~~ Hartmann and Hassel-
50 horn (2008) provides a scenario-based measure of HPT. It was validated with German
51 ~~secondary students, it has been adapted for use in the Netherlands and employed in~~
52 ~~comparative studies of historical reasoning~~Grade 10 students (Hartmann & Hasselhorn,
53 2008) and subsequently translated and studied with Dutch students aged 10–17 (Huijgen,
54 van Boxtel, van de Grift, & Holthuis, 2014, 2017). This evidence base is promising
55 but remains concentrated in two language contexts and a small number of studies.

56 The instrument operationalizes HPT through a scenario-based approach in which
57 students read about a historically situated dilemma and respond to nine Likert-scaled
58 items ~~capturing three developmental levels: presentism/othering of people in three~~
59 interpretive response categories: present-oriented perspective (POP), ~~recognition of~~
60 ~~the agent's social role and situational context~~ role of the historical agent (ROA), and
61 historical contextualization (CONT). ~~This structure reflects a theoretical progression~~

~~from naive~~ POP and CONT represent the clearest contrast between present-centered judgment ~~toward increasingly sophisticated historical reasoning and contextual reasoning;~~ ROA's position as an intermediate level remains theoretically less certain (Hartmann & Hasselhorn, 2008; Lee & Ashby, 2001).

The present study reports ~~the first a~~ Czech adaptation and psychometric evaluation of the Hartmann and Hasselhorn HPT instrument. Two considerations motivate this work. First, no validated Czech-language HPT instrument ~~currently exists, despite growing interest in historical thinking assessment within Czech education reform .~~ The Czech adaptation ~~addresses this gap and provides the research community with a tool for cross-cultural comparisons of HPT development~~ was identified. Czech initiatives have developed digital ~~historical-literacy tasks and measures of epistemic beliefs about history~~ (Ripka, Sýkorová, Münich, & Chvojka, 2024; Činátl, Najbert, & Ripka, 2021); ~~the present adaptation provides a candidate instrument for further Czech validation and future cross-national equivalence studies.~~

Second, the instrument's scenario content raises a specific fairness question. ~~The scenario describes a young man in the Weimar Republic deliberating whether to join an extremist political movement. Items that require students to contextualize this figure's reasoning Hannes discusses Germany's crisis and the 1930 election; students judge how well nine explanations fit his situation, including whether he could vote for an anti-democratic party such as the NSDAP. These judgments~~ could, in principle, function differently for students whose political attitudes align with the scenario's ideological content. Students with right-authoritarian sympathies might find it cognitively easier to endorse

contextualizing responses—not because they are reasoning historically, but because the responses resonate with their own attitudes. If present, such a pattern would constitute construct-irrelevant variance (CIV): systematic score variation attributable to a factor extraneous to the target construct (American Educational Research Association et al., 2014; Messick, 1995).

This concern is not merely theoretical. Research on identity-protective cognition demonstrates that individuals process information through the lens of pre-existing beliefs (Jost, Glaser, Kruglanski, & Sulloway, 2003; Kahan, Peters, Dawson, & Slovic, 2017). Recent work has documented how ideological content can infiltrate competence judgments in history assessment, whether through teacher scoring bias in open-ended narrative assessment (Alvén, 2019) or through construct comparability challenges in cross-cultural settings (Badham, Meadows, & Baird, 2025). The question central to the present study is whether the HPT instrument’s psychometric properties—its factor structure, item parameters, and score interpretations—hold equivalently across students with different political attitudes, or whether politically charged content introduces measurement bias.

The study contributes to the literature in three ways: (a) it ~~demonstrates a transferable~~ applies a possible methodological approach for evaluating whether attitudinally charged assessment content introduces construct-irrelevant variance, (b) it provides DIF and ~~measurement invariance evidence for a fairness-relevant grouping variable—political ideology—that has not previously been examined in HPT measurement~~ measurement invariance evidence across political-ideology groups, and (c) it ~~reports the first published Czech validation of the Hartmann and Hasselhorn HPT instrument, extending the~~ adds Czech

evidence to the small cross-cultural evidence base for this measure. To establish this evidence, we conducted confirmatory factor analysis (CFA), reliability analysis, intra-class correlation estimation, IRT-based differential item functioning (DIF) analysis, and multi-group CFA for measurement invariance testing.

2. Literature Review

2.1 Historical Perspective Taking and Its Measurement

HPT refers to the ability to understand past actors' decisions by reconstructing the historical context in which they operated (Hartmann & Hasselhorn, 2008; Lee & Ashby, 2001). HPT is a second-order (procedural or disciplinary) concept—as distinct from first-order (substantive) concepts such as “revolution” or “empire,” it concerns the meta-level reasoning processes historians use to make sense of the past (Seixas & Peck, 2004). The concept has roots in British traditions that progressively refined the construct from historical empathy (*Children's concepts of empathy and understanding in history*, 1987; Shemilt, 1984) toward rational reconstruction (Lee & Ashby, 2001). It has been situated within broader frameworks of historical reasoning that encompass evidence use, argumentation, and substantive concept deployment—the application of domain-specific content knowledge about events, causes, and actors (van Boxtel & van Drie, 2018; van Drie & van Boxtel, 2008; VanSledright, 2014). The literature variously labels this capacity “historical empathy,” “perspective taking,” or “rational reconstruction.” We follow Hartmann and Hasselhorn (2008) in adopting the term “perspective taking” within the rationalist tradition of ~~which emphasises~~ Lee and Ashby (2001), which emphasizes cognitive reconstruction

of context rather than affective identification with past actors (see Barton & Levstik, 2004, for a contrasting view that emphasises affective engagement alongside cognitive reconstruction; see also ~~for a more recent treatment~~ Endacott, 2014; Endacott & Brooks, 2013, for treatments of the empathy/perspective-taking distinction). The present instrument therefore operationalises cognitive perspective taking, not the fuller cognitive-affective construct of historical empathy. Across these frameworks, HPT is understood as a cognitively demanding, knowledge-dependent competence—not a disposition but a disciplinary achievement requiring both domain-specific knowledge and the capacity to deploy that knowledge in reconstructing past perspectives (Chapman, 2021; VanSledright, 2014; Wineburg, 2001). Distinguishing genuine disciplinary reasoning from attitude-driven responding is a measurement challenge that extends beyond HPT to epistemic cognition in history more broadly (Maggioni, VanSledright, & Alexander, 2009).

~~operationalised HPT into a developmental model with three levels:~~ Hartmann and Hasselhorn (2008) operationalized HPT using three response categories: ~~presentism~~present-oriented perspective (POP), where students judge historical actors by contemporary standards; ~~recognition role of the historical agent's role~~ (ROA), where students ~~acknowledge the agent's social position without full contextual reconstruction~~explain action through familiar or stereotyped social roles; and *contextualization* (CONT), where students situate decisions within the specific historical conditions that rendered them intelligible. ~~This developmental progression from present-centred judgment toward contextualisation~~The POP–CONT continuum from present-centered judgment toward contextualization has empirical parallels in earlier work on students' historical understanding (Lee & Shemilt,

2003). The original published validation study ($n = 170$ German students, Grade 10) ~~recovered three levels through principal component analysis~~; Hartmann & Hasselhorn, 2008) placed raw POP and CONT at opposing poles of one principal component and ROA on a second, less clearly interpreted component (explained variance = 51%), ~~and students classified at the highest level~~; a separate latent-class analysis identified three student classes. Students classified in the highest class had significantly better history grades; however, PCA is an exploratory technique, and the present study ~~provides the first confirmatory factor analytic test of the three-level structure. The instrument was extended by~~ tests a correlated three-factor representation rather than reproducing the original PCA dimensions. Huijgen et al. (2014) ~~and~~ replicated the two-dimensional pattern for the Nazi-party scenario with Dutch students, but transfer of the item-rating approach to a slavery scenario produced weaker reliability and validity evidence. Huijgen et al. (2017) ~~who confirmed the core theoretical structure while highlighting the difficulty of contextualization for most students~~ subsequently combined the closed item ratings with think-alouds and documented how students interpreted and sometimes misunderstood the response options. Thus, successful reuse of the Nazi scenario does not establish that the format transfers unchanged across topics or elicits the intended process.

The difficulty of measuring HPT stems from its inherently interpretive character. Unlike factual recall, HPT cannot be ~~assessed through simple multiple-choice items with single correct answers~~ adequately inferred from decontextualized factual-recall items alone (Breakstone, 2014; Smith, 2017). Scenario-based instruments address this challenge by presenting contextual information and requiring students to evaluate statements reflect-

ing different levels of historical reasoning (Hartmann & Hasselhorn, 2008; Seixas, Gibson, & Ercikan, 2015). However, this approach introduces a measurement tension: the richer the historical context, the more opportunity for construct-irrelevant factors—including attitudes toward that content—to influence responses.

2.2 Cross-Cultural Test Adaptation

The adaptation of educational and psychological instruments across languages and cultures requires systematic procedures to ensure that the adapted version measures the same construct as the original (Hambleton et al., 2005; International Test Commission, 2017). The International Test Commission (ITC) guidelines emphasize that adaptation involves more than translation: it requires attention to construct equivalence, cultural appropriateness, and technical quality of the adapted version (International Test Commission, 2017). For instruments used in history education, cross-cultural adaptation faces the additional challenge that historical content may carry different resonances across national contexts. A scenario about the Weimar Republic, originally designed for German students, intersects differently with Czech historical memory—while subsequent events shape contemporary Czech memory of the interwar period, the scenario centres on Weimar-era de-liberation (early 1930s), which predates the 1938 Munich Agreement and cession of the Sudetenland by approximately ~~five~~eight years. Nevertheless, the broader period connects directly to the Nazi occupation of 1939–1945 in Czech collective memory.

In Czech lower-secondary education, history is prescribed within the compulsory People and Society curriculum area for Grades 6–9. The framework includes the 1920s and 1930s,

192 fascism, Nazism, totalitarianism, and work with historical sources, but does not define
193 HPT as a separate competence (Ministerstvo školství, mládeže a tělovýchovy, 2023, pp. 54–
194 58, 140, 143). School curricula broadly align with this framework, although curriculum
195 documents describe intended rather than enacted provision (Kaleta, 2025). Participating
196 teachers reported that students had encountered the relevant interwar content; this does
197 not establish equivalent prior instruction across schools.

198 This documentary account applies specifically to the lower-secondary subsample. The
199 majority of participants were enrolled in upper-secondary education, for which this study
200 did not conduct an equivalent curriculum- document audit. Eligibility at both levels
201 required teacher confirmation that students had completed the interwar unit, but that
202 confirmation is not evidence of uniform curricular coverage or instruction.

203 Cross-cultural validation studies should demonstrate that the adapted instrument pre-
204 serves the factor structure, reliability, and validity relationships of the original (Ham-
205 bleton et al., 2005). At minimum, this requires confirming the theorized factor structure
206 through CFA, reporting reliability estimates appropriate for the item characteristics, and
207 providing convergent validity evidence through expected correlations with related con-
208 structs. When the instrument will be used to compare groups or predict outcomes, mea-
209 surement invariance testing becomes essential (American Educational Research Associa-
210 tion et al., 2014).

2.3 Construct-Irrelevant Variance in Educational Assessment

The unified validity framework identifies construct-irrelevant variance as a principal threat to score interpretation (American Educational Research Association et al., 2014; Messick, 1995). CIV occurs when systematic variance in test scores is attributable to factors extraneous to the target construct. When assessment content intersects with respondents' attitudes or social identities, CIV can arise through attitude-consistent responding: students whose beliefs align with test content may find certain response options more natural or appealing, inflating their scores without corresponding gains in the assessed competence (Messick, 1995).

For the HPT instrument specifically, the concern is that the Weimar Republic scenario creates a *congruence pathway*—a mechanism whereby attitudinal similarity to item content lowers the cognitive effort required to endorse a response, inflating scores independently of competence. Consider a contextualization item such as “His decision was understandable given the political and economic turmoil of the Weimar Republic.” A student who endorses this statement because she has reconstructed the historical context produces a valid HPT response. But a student who endorses it because the sentiment resonates with her own political sympathies—not because she has engaged in historical reasoning—produces an inflated score. This would represent construct contamination in Messick's (1995) terms—a systematic source of score variance unrelated to HPT competence. Evidence of internal structure, including measurement invariance across relevant subgroups, constitutes essential validity evidence when scores are used to compare groups or when the construct definition requires demonstrating that extraneous factors do not influence

scores (American Educational Research Association et al., 2014).

An alternative theoretical prediction deserves mention. Research on authoritarian cognition suggests that high authoritarianism is associated with reduced integrative complexity—the cognitive capacity to hold and reconcile multiple conflicting perspectives simultaneously (Duckitt, 2001; Suedfeld & Tetlock, 1977). Because HPT requires precisely this kind of integrative capacity, authoritarian students might perform *worse*, not better. The direction of any bias, if present, was genuinely uncertain. Additionally, dual-process theories of cognition (Evans & Stanovich, 2013) suggest that structured assessment formats may channel deliberative processing, overriding the automatic attitude-retrieval processes that would produce attitude-consistent responding. If so, the assessment format itself may buffer against attitudinal interference.

The construct boundary is therefore consequential. Under broader accounts of historical empathy, affective engagement and morally responsive evaluation can be target-relevant aspects of understanding a past actor (Barton & Levstik, 2004; Endacott & Brooks, 2013). Under the narrower cognitive definition operationalized here, target variance consists of reconstructing context and integrating historically relevant considerations. Moral reaction or affect is not itself scored, and endorsement driven by option familiarity, task demands, or attitude congruence is construct-irrelevant when it substitutes for that reconstruction. The rating data alone cannot determine which process produced an endorsement.

2.4 DIF and Measurement Invariance as Fairness Evidence

Differential item functioning and measurement invariance testing provide complementary approaches to evaluating measurement fairness (Chen, 2007; Cheung & Rensvold, 2002; Finch, 2005). DIF analysis examines whether individual items function differently across groups after controlling for the latent trait, identifying specific items that may be biased. Measurement invariance testing through multi-group CFA examines whether the overall measurement model holds across groups at three progressively restrictive levels: *configural invariance* (the same factors exist in both groups), *metric invariance* (items relate equally strongly to the factors in both groups), and *scalar invariance* (the scale origin is identical, so mean scores can be directly compared across groups).

For ordinal items, the weighted least squares mean and variance adjusted (WLSMV) estimator—a robust estimation method designed for categorical data—is recommended for CFA (Flora & Curran, 2004). Because ordinal items have discrete response categories rather than continuous scales, threshold-based invariance testing (which models the boundaries between response categories) replaces intercept-based approaches. The conventional criteria for evaluating invariance steps use changes in fit indices rather than chi-square difference tests, which are sensitive to sample size: $\Delta CFI \leq .01$ and $\Delta RMSEA \leq .015$ indicate that the more constrained model does not fit meaningfully worse (Chen, 2007; Cheung & Rensvold, 2002).

In the context of HPT measurement, establishing scalar invariance across ideology groups would mean that the factor structure, loadings, and item thresholds are equivalent for students with different political attitudes—a necessary condition for concluding that ob-

served score differences (or the lack thereof) reflect true construct differences rather than measurement artifacts.

2.5 The Present Study

This study addresses two interrelated measurement questions:

1. ~~Does~~ What psychometric evidence does the Czech adaptation of the Hartmann and Hasselhorn HPT instrument ~~demonstrate acceptable psychometric properties, including the theorized~~ provide, including evidence concerning a correlated three-factor ~~structure, adequate~~ representation, score reliability, and ~~convergent validity~~ convergence with historical knowledge?

2. ~~Does the instrument function fairly across students with different political attitudes, as evidenced by the absence of DIF and by scalar measurement invariance across ideology groups~~ What evidence do DIF, measurement-invariance, and regression analyses provide about whether scores are affected by students' political attitudes?

~~We hypothesized that the three-factor structure would be confirmed and that the instrument would demonstrate measurement invariance across ideology groups. As convergent validity evidence, we expected historical knowledge to correlate positively with HPT scores, consistent with the theoretical conceptualization of HPT as a knowledge-dependent competence~~ The study and analysis plan were preregistered. The present measurement-focused organization, the six-item contextualization composite as the primary outcome, and several analyses were adopted after registration, and not all registered analyses were implemented. Table S5 maps the registered hypotheses,

omissions, and post-registration additions; findings from the latter are treated as
exploratory or descriptive rather than as preregistered tests.

3. Method

3.1 Adaptation Process

The Czech HPT instrument was developed through a systematic translation and adaptation process following ITC guidelines (International Test Commission, 2017). Two bilingual Czech-English experts conducted independent forward translations from the ~~published English version of the instrument~~German instrument reported in Hartmann (2008); wording was cross-checked against the published English validation article (Hartmann & Hasselhorn, 2008). An independent backward translation was produced by a third bilingual expert unfamiliar with the original. Five Czech history educators at the secondary level reviewed the adapted items for conceptual equivalence, age-appropriateness, and curricular alignment. No items required substantive modification; minor wording adjustments were made to two items for naturalness in Czech. Reviewers ~~confirmed that the Weimar Republic scenario content was familiar~~judged the scenario
content accessible to Czech secondary students ~~through the national history curriculum~~
~~, given the direct connections between the interwar period and subsequent Czech~~
~~historical experience (the Munich Agreement, the cession of the Sudetenland, and the~~
~~Nazi occupation)~~given the curriculum coverage described in Section 2.2. This was
expert-review evidence, not direct evidence from student interviews.

The adaptation preserved the ~~original instrument's structure: the~~ four-point response

format (1 = *strongly disagree* to 4 = *strongly agree*), the ~~three-factor-item-structure~~ three
item categories (three items per ~~subscale~~category), and ~~the~~ reverse-scoring of POP
items ~~were retained unchanged from the validated German version~~ from the German
source. All deviations from the original were documented transparently in the OSF
repository (<https://doi.org/10.17605/OSF.IO/YNG37>). The ~~scenario narrative was~~
author separately cross-checked the scenario against the Czech history curriculum ~~to~~
~~confirm that all contextual references were accessible to the target population, but no~~
direct student response-process study was conducted.

No cognitive pilot study (e.g., think-aloud interviews) was conducted prior to main data
collection, which constitutes a limitation of the adaptation process (see Section 5.6).

3.2 Participants

The sample comprised 293 students from 20 classrooms across 10 schools in the Czech
Republic, spanning lower-secondary (International Standard Classification of Education
[ISCED] level 2, approximately ages 11–15; $n = 87$, 29.7%) and upper-secondary (ISCED 3,
approximately ages 15–19; $n = 206$, 70.3%) levels. The inclusion criterion was completion
of the curricular unit on the interwar period. Schools were drawn from five Czech regions
(Praha: 42.7%; Královéhradecký: 34.8%; Zlínský: 11.3%; Jihomoravský: 8.2%; Ústecký:
3.1%) and three funding categories (public: 90.8%; private: 6.5%; church-affiliated: 2.7%).
Gender composition was 192 male (65.5%), 88 female (30.0%), and 13 identifying as an-
other gender (4.4%). Students' most recent history grade averaged 1.71 ($SD = 0.92$; Czech
scale: 1 = excellent, 5 = failing).

Schools were recruited through convenience sampling; the sample over-represents upper-secondary students relative to the national age cohort composition and should not be treated as a probability sample of Czech adolescents. Sample size was determined by an a priori power analysis specified in the preregistered protocol (OSF: <https://doi.org/10.17605/OSF.IO/YNG37>), targeting 80% power to detect standardized ideology effects of $\beta \geq .15$ in a two-level multilevel ~~model~~regression; the achieved $N = 285-293$ met ~~this target.~~that regression target. The calculation did not establish power for the post-registration DIF or MG-CFA analyses. Within each school, all participating students present on the day of administration completed the instrument battery during a regular class period (approximately 30–40 minutes). Teachers were present during administration but did not intervene in students' responses.

The sample's right-authoritarian attitude median was 2.33 on the five-point scale, and the historical-knowledge median was 3 of 6; fuller distributions appear in Section 4.4. No defensible student-, family-, or school-level socioeconomic indicator was collected. Consequently, the sample's socioeconomic composition and any associated selection bias cannot be characterized.

3.3 Instruments

3.3.1 Historical Perspective Taking (HPT) Students completed the Czech adaptation of the scenario-based HPT instrument from the German dissertation (Hartmann, 2008), with the published English account used as supporting validation evidence (Hartmann & Hasselhorn, 2008). They read a narrative ~~about a young man in the Weimar Republic~~

~~considering joining an extremist political movement, then responded to nine items~~
in which Hannes discusses Germany's crisis and the 1930 election, then judged nine
explanations of his situation, including whether he could vote for an anti-democratic
party such as the NSDAP, on a 4-point Likert scale. Three items each tapped ~~presentism~~
present-oriented perspective (POP1–POP3), ~~recognition of the agent's role~~ role of the
historical agent (ROA1–ROA3), and historical contextualization (CONT1–CONT3). POP
 items capture present-centered ~~moral~~ judgments of the historical figure ~~(e.g., evaluating~~
~~his decision by contemporary standards)~~; these are reverse-scored (5 – POP, yield-
 ing a reversed range of 1–4) so that higher scores indicate ~~more contextualized~~ less
present-oriented reasoning. ROA items ~~reflect awareness of the agent's social position~~
~~and circumstances without full contextual integration~~ explain action through familiar or
stereotyped social roles. CONT items ~~represent the most sophisticated level, requiring~~
require students to situate the historical figure's deliberation within the specific politi-
 cal, economic, and social conditions of the Weimar Republic. The primary ~~composite~~
~~was HPT_CTX6,~~ six-item contextualization composite was the mean of reversed POP
~~and CONT items.~~ HPT_TOT9 (adding ROA) served present-oriented perspective and
contextualization scores and therefore represents the continuum between these two
response categories. The nine-item total score (adding the theoretically less certain
role-of-the-agent category) served only as a secondary sensitivity outcome.

3.3.2 Right-Authoritarian Ideology (FR-LF mini) A six-item short form of the Frage-
 bogen Rechtsextremer Einstellungen—Leipziger Form (Questionnaire on Right-Wing Ex-
 tremist Attitudes—Leipzig Form; Decker, Kiess, & Brähler, 2013), comprising acceptance

of right-wing dictatorship (RD; 3 items) and National Socialist sympathy/relativization (NS; 3 items) facets, rated on a 5-point scale. The NS facet is theoretically the most direct operationalization of potential content congruence with the Weimar scenario.

3.3.3 Authoritarianism (KSA-3) The nine-item Kurzskala Autoritarismus (Short Scale of Authoritarianism; Beierlein, Asbrock, Kauff, & Schmidt, 2014), measuring authoritarian aggression, submission, and conventionalism on a 5-point scale. This instrument was used for a supplementary measurement invariance check using an alternative ideology operationalization.

3.3.4 Historical Knowledge (KN) Six multiple-choice items covering content relevant to the HPT scenario, scored dichotomously (sum score 0–6). This measure served as a convergent validity criterion, given the theoretical expectation that HPT is a knowledge-dependent competence.

3.3.5 Social Desirability (SDR-5) The five-item short form (Hays, Hayashi, & Stewart, 1989), rated on a 5-point scale with items SDR2–SDR4 reverse-coded. Included to evaluate whether HPT scores reflect impression management rather than genuine reasoning.

The source studies for the FR-LF included German respondents aged 14 years and older, whereas KSA-3 was developed for German-speaking adults aged 18 years and older and SDR-5 with adult medical outpatients and older adults. None therefore constitutes direct Czech adolescent validation; suitability rests on expert review and the sample-specific evidence reported here and in Table S4.

3.4 Analytic Approach

All analyses were conducted in R (version 4.4.2) using lavaan for CFA, mirt for IRT-based DIF analysis, psych for reliability estimation, and lme4/performance for intraclass correlations.

Factor structure. CFA compared one-factor (general HPT), two-factor (POP+CONT vs. ROA), and three-factor (POP, ROA, CONT) models using the WLSMV estimator, appropriate for ordinal item responses (Flora & Curran, 2004). CFA fit indices summarize how well a hypothesized factor structure reproduces the observed correlations among items; multiple indices are reported because each captures a different aspect of misfit. Model evaluation used CFI ($\geq .95$ acceptable, $\geq .97$ good), the Tucker–Lewis Index (TLI; $\geq .95$), the Root Mean Square Error of Approximation (RMSEA; $\leq .08$ acceptable, $\leq .05$ good), and the Standardized Root Mean Square Residual (SRMR; $\leq .08$).

Reliability. Internal consistency was assessed ~~via~~ using raw and polychoric Cronbach's α ~~(raw and based on polychoric correlations) and~~ McDonald's ω , which does not assume tau-equivalence (the assumption that all items relate equally strongly to the latent factor) and is generally preferred for congeneric measures (measures whose items may differ in how strongly they relate to the factor;) and KR-20 for the dichotomous knowledge items. Reliability of the combined ideology score was estimated with the Mosier formula. Full methods and estimates are reported in Table S3.

Intraclass correlations. We computed intraclass correlation coefficients (ICCs) at the school and classroom levels to ~~characterise~~ characterize the multilevel structure of the

data and to evaluate whether clustering warranted multilevel approaches.

Clustered data handling. The ~~DIF and~~ substantive regressions used random intercepts for the nested data. The DIF, MG-CFA analyses treat, and zero-order correlation analyses treated observations as independent, ~~which is standard practice in applied DIF and measurement invariance research. To evaluate whether ignoring the nested structure introduces meaningful bias, design effects (DEFF—a multiplier indicating how much clustering inflates standard errors relative to a simple random sample) were computed post hoc. A design effect of 1.0 means clustering has no impact on precision; values approaching 2.0 indicate modest inflation of standard errors that can usually be tolerated. The primary composite (HPT_CTX6) had DEFF = 1.56, within the conventional ≤ 2.0 threshold. The secondary composite (HPT_TOT9) had DEFF = 2.16, marginally above this threshold; consequently, inferential claims are anchored primarily to HPT_CTX6, and HPT_TOT9 results should be interpreted with this caveat.~~¹ . Composite score ICCs do not establish that item thresholds, loadings, or their standard errors are unaffected by clustering; therefore, uncertainty for these analyses may be understated. The zero-order correlation intervals are reported as descriptive, unadjusted intervals rather than cluster-robust inference.

Missing data and score construction. Composite scores required at least two answered items per three-item HPT subscale, four of six FR-LF items, seven of nine KSA-3 items, and four of five SDR-5 items; available responses were averaged. Missing knowledge

¹ $DEFF = 1 + (\bar{n}_c - 1) \times ICC$, where $\bar{n}_c$ is the average cluster size (approximately 14.7 students per school). The TYPE=COMPLEX option in lavaan was not applied because the DIF analysis in mirt does not support clustered standard errors, and applying different corrections to DIF and MG-CFA would make the two sets of results less directly comparable.

responses were scored as incorrect, following the original analysis scripts. Reliability estimates used listwise-complete item sets, while correlations used pairwise-complete composite scores. These rules explain the analysis-specific sample sizes reported in the tables and supplement.

Differential item functioning. Low and High ideology groups were formed post-registration using tertile splits on a composite ideology score (mean of z-scored FR-LF and KSA-3), with the middle tertile excluded to maximize contrast (Low $n \approx = 96$ per group). This extreme-group design follows established practice in DIF research. A constrained; High $n = 95$). A constrained unidimensional multi-group graded response model (GRM item discrimination and threshold-category-intercept parameters for each item by group are reported in Table S1 in the OSF supplementary materials) using a sequential item-release design tested each item for uniform DIF (items are systematically easier or harder for one group at all ability levels) and non-uniform DIF (the group difference reverses depending on ability level) omnibus DIF by jointly freeing its discrimination and category intercepts, with Bonferroni correction ($\alpha = .01$). The baseline model constrains all nine items to equality across groups; each test then frees the discrimination and threshold parameters-category intercepts for the focal item, yielding a likelihood-ratio comparison against the fully constrained baseline (an all-others-as-anchor strategy). This approach assumes that the remaining eight items are DIF-free when testing each focal item; iterative purification was not applied, which means that if multiple anchor items exhibited small DIF in the same direction, anchor contamination could mask detection. focal-item detection could be affected. Because the CFA favoured

correlated dimensions, the unidimensional GRM is a pragmatic sensitivity model rather than a fully aligned measurement model; multidimensionality or local dependence may affect the item tests.

Sensitivity of DIF detection. With approximately 96 respondents per ideology group, the likelihood-ratio DIF tests at Bonferroni-corrected $\alpha = .01$ (9 tests) are powered primarily to detect moderate-to-large parameter differences ($\Delta\beta \geq 0.5$ logits on discrimination or threshold parameters). Weak DIF effects ($\Delta\beta < 0.5$ logits) cannot be confidently excluded. This sensitivity threshold is consistent with the sample size requirements documented for IRT-based DIF methods and nine Bonferroni-adjusted tests, statistical power was limited. No design-specific simulation was conducted, so the minimum detectable DIF magnitude is unknown. The extreme-group design maximizes statistical contrast between ideology subgroups increases ideological contrast at the cost of reduced per-group sample size—a standard trade-off in DIF research when the grouping variable is continuous excluding the middle tertile and reducing group size.

Multi-group CFA. Stepwise MG-CFA tested configural, metric, and scalar invariance across ideology groups using the three-factor model with WLSMV estimation. Invariance was evaluated using $\Delta CFI \leq .01$ and $\Delta RMSEA \leq .015$ criteria (Chen, 2007; Cheung & Rensvold, 2002).

Substantive models. Random-intercept linear models predicted each standardized HPT outcome from standardized FR-LF, KSA-3, historical-knowledge, and SDR-5 scores entered together, with intercepts for school and classroom. Thus, each reported ideology coefficient is mutually adjusted for the other ideology measure, knowledge, social

desirability, and the nested grouping structure.

Convergent validity. Descriptive Pearson correlations between HPT composites and historical knowledge provided convergent validity evidence; Fisher-transformed intervals did not adjust for clustering.

Supplementary invariance check. To verify that invariance findings were not an artifact of the composite ideology definition, MG-CFA was also conducted using NS-only grouping (the facet most directly content-congruent with the Weimar scenario).

4. Results

The Results follow the two measurement questions in sequence. Sections 4.1–4.5 assess factor structure, score precision, distributions, and clustering; Sections 4.6–4.8 examine item and model-level fairness; Sections 4.9–4.10 then report the focal knowledge and ideology relationships.

4.1 Factor Structure

CFA supported the three-factor model (POP, ROA, CONT) as best-fitting, outperforming the two-factor and one-factor alternatives (Table 1). The three-factor model showed excellent global fit: CFI = .999, TLI = .998, RMSEA = .011 (90% CI [.000, .051]), SRMR = .048. The two-factor model combining POP and CONT items on a single factor showed acceptable but inferior fit (CFI = .956, RMSEA = .057), while the one-factor model fit poorly (CFI = .926, RMSEA = .073).

Table 1. *CFA Fit Indices for HPT Factor Models (WLSMV Estimator)*

Model	χ^2	df	CFI	TLI	RMSEA	90% CI	SRMR
1-Factor	66.68	27	.926	.902	.073	[.051, .095]	.075
2-Factor (POP+CONT, ROA)	49.48	26	.956	.940	.057	[.032, .081]	.066
3-Factor (POP, ROA, CONT)	24.78	24	.999	.998	.011	[.000, .051]	.048

Note. WLSMV = weighted least squares mean and variance adjusted. All items treated as ordered categorical. The 2-factor model groups POP and CONT items on a single factor with ROA on a separate factor. Chi-square values are mean-and-variance adjusted (WLSMV). Nested model comparisons require the DIFFTEST procedure rather than standard chi-square difference tests. All $p < .001$ except the three-factor model ($p = .418$).

The near-perfect global fit of the three-factor model reflects the small item set (nine items in three just-identified three-indicator factors—each subscale has exactly as many free parameters as observed statistics, leaving no room for misfit within subscales) and should be interpreted alongside local fit diagnostics: the largest modification index was 7.35 (POP1–ROA3 residual covariance; below the threshold of 10), and standardized residual correlations were small ($|r| \leq .13$). With only $df = 24$ for the three-factor model (9 items, 3 factors, 24 free parameters against 45 observed statistics), there is minimal room for three indicators per factor, there is limited opportunity to reveal within-factor misfit; the near-perfect CFI is therefore a structural consequence of the low- df specification rather than evidence of an unusually well-fitting model. ~~These results confirm that the Czech adaptation preserves the theorized three-level structure of the original German instrument~~The Czech data are compatible with the correlated three-factor specification, but this is not a test of

cross-national equivalence.

Inter-factor correlations from the three-factor CFA were: POP–ROA $\Phi = .377$, POP–CONT $\Phi = .509$, ROA–CONT $\Phi = .631$. These moderate-to-strong correlations ~~indicate meaningful associations among the HPT dimensions, with ROA and CONT showing the strongest link. Discriminant validity is supported because all inter-factor correlations are below .85, confirming that the three subscales capture related but distinct aspects of historical perspective taking~~ and the better fit of the three-factor model provide internal-structure evidence that the item sets are distinguishable in this sample. They do not by themselves establish discriminant validity. Stronger evidence would require showing that the subscales relate differentially to external criteria in theoretically expected ways.

A supplementary bifactor model (one general factor plus three specific factors, in which each item loads on both the general factor and its subscale-specific factor) fit well ($\chi^2 = 14.70$, $df = 18$, $p = .682$) but did not meaningfully improve over the correlated three-factor model ($\Delta CFI = .001$). The low df (18) results from the additional cross-loadings consuming degrees of freedom, and the comparison with the correlated three-factor model ($df = 24$) should be treated cautiously given the different parameterisations. CONT items loaded primarily on the general factor (.47–.63), while POP and ROA retained meaningful specific factor variance, supporting subscale-level interpretation.

4.2 Standardized Factor Loadings

Table 2 presents the standardized loadings from the three-factor CFA. CONT items loaded most strongly (.61–.72), followed by POP items (.30–.74) and ROA items (.30–.67). POP2 showed the weakest loading (.30), at the conventional minimum threshold for item retention; it was retained to preserve fidelity to the original instrument structure. The range of loadings across subscales reflects the heterogeneity within each three-item set and is consistent with patterns reported in prior validations of short-form instruments.

Table 2. *Summary of Standardized Factor Loadings for the Three-Factor HPT Model*

Subscale	Loading range	Weakest item
Present-oriented perspective	.30–.74	POP2 (.30)
Role of historical agent	.30–.67	ROA2 (.30)
Historical contextualization	.61–.72	CONT2 (.61)

Note. POP items are reverse-scored (higher = more contextualized reasoning). All loadings were significant at $p < .001$. WLSMV estimator with items treated as ordered categorical. Full item estimates, standard errors, and confidence intervals are reported in Table S2.

4.3 Reliability and Internal Consistency

Table 3 presents reliability coefficients for each HPT subscale. Raw summarizes reliability for the primary study scores. Across HPT subscales, raw Cronbach’s α values ranged from .47 (POP) to .64 (CONT); polychoric α (appropriate for ordinal items)

ranged from .53 to .69; and McDonald's ω ranged from .55 to .69. CONT demonstrated the strongest reliability, consistent with its highest average factor loadings from .54 to .70. Full estimates for all HPT and predictor scores are in Table S3.

Table 3. Reliability Coefficients Summary for HPT Subscales Primary Study Scores

Score	Items/components	Reliability	Method
HPT composite	6	.66	ω total
HPT total	9	.70	ω total
Combined ideology	2 scales	.80	Mosier

Note. POP items are reverse-scored. The Mosier estimate has a 95% bootstrap CI [.74, .84]. Full subscale and predictor alpha, polychoric-alpha, omega, and interval estimates are reported in Table S3.

Reliability was modest and constrains individual interpretation. FR-LF had α = Cronbach's alpha; ω .67, KSA-3 had α = McDonald's omegatotal. Mean r_{ii} .72, and the combined ideology score had estimated reliability .80, 95% CI [.74, .84]. Knowledge had KR-20 = mean inter-item correlation.

These values are modest by conventional standards but structurally constrained by the three-item design: with $k = 3$, even moderate inter-item correlations (.20-.35) yield .55, and SDR-5 was weak (α in this range. The Spearman-Brown formula (which estimates how reliability would improve if subscale length were doubled) projects $\alpha = .64$ -.79 for six-item subscales, though this assumes tau-equivalence—an assumption violated

when loadings range from .30 to .74 (Table 2)—so the projections should be treated as approximate upper bounds. McDonald's ω (.55–.69) is more appropriate given unequal loadings (Table 2). The knowledge–HPT correlations reported below ($r = .22$ –.34) demonstrate that the instrument carries genuine psychometric signal despite modest subscale reliability. Correcting the ideology–HPT correlation for measurement unreliability (disattenuation—an adjustment that estimates what the correlation would be if both measures had perfect reliability) yielded a disattenuated $r = .07$, indicating that measurement error does not mask a meaningful ideology association. However, reliability for the composite ideology score (the mean of z-scored FR-LF and KSA-3) was not separately estimated, which limits the precision of this disattenuated estimate.

4.4 Descriptive Statistics

Table 4 presents descriptive statistics. HPT scores followed the expected developmental ordering (POP_rev > ROA > CONT), with contextualization producing mean-score ordering: reversed present-oriented perspective scores were highest, followed by role-of-the-agent and contextualization scores. Contextualization produced the lowest scores ($M = 2.71$). Ideology scores were below the scale midpoint (i.e., this ordering may reflect item endorsement or difficulty and is not developmental evidence. Right-authoritarian attitude scores (FR-LF) and authoritarianism scores (KSA-3) were below their scale midpoints ($M = 2.48$; KSA: $M = 2.86$), reflecting the rarity of strong, respectively). The right-authoritarian endorsement in mainstream adolescent samples: attitude median was 2.33 (IQR [2.00, 3.00]) and 51 of 284 students (18.0%) scored above its midpoint. The authoritarianism median was 2.89 (IQR [2.53, 3.22]) and 106 of 283

students (37.5%) scored above its midpoint. On the mean of the two raw scales, 71 of 283 students (25.1%) scored above 3. Historical knowledge had a median of 3 (IQR [2, 4]), with 4.1% at the floor and 8.9% at the ceiling. No defensible student- or school-level socioeconomic measure was available.

Table 4. *Descriptive Statistics for Study Variables*

Variable	<i>n</i>	<i>M</i>	<i>SD</i>	Min	Max
Present-oriented perspective (reversed)	287	2.97	0.64	1.33	4.00
Role of the historical agent	287	2.80	0.68	1.00	4.00
Historical contextualization	287	2.71	0.74	1.00	4.00
Six-item contextualization composite	287	2.84	0.55	1.33	4.00
Nine-item HPT total score	287	2.83	0.49	1.67	3.89
Historical knowledge (0–6)	293	3.04	1.62	0.00	6.00
Dictatorship acceptance	285	2.54	0.88	1.00	5.00
Nazi-crime relativization	286	2.43	0.89	1.00	5.00
Right-authoritarian attitudes	284	2.48	0.75	1.00	5.00
Authoritarianism	283	2.86	0.62	1.00	4.60
Social desirability	283	3.02	0.63	1.00	4.60

Note. HPT items scored on a 1–4 scale; POP items reverse-coded. FR-LF and KSA-3 scored on a 1–5 scale. KN scored as sum of correct responses (0–6).

4.5 Intraclass Correlations

Intraclass correlations indicated meaningful school-level clustering for most HPT outcomes (HPT_CTX6six-item contextualization composite: $ICC_{\text{school}} = .041$; HPT_TOT9nine-item total score: $ICC_{\text{school}} = .085$; ROArole of the historical agent: $ICC_{\text{school}} = .102$), consistent with between-school differences in instructional emphasis and student composition. Classroom-level variance was near zero for the primary composite but non-negligible for CONTcontextualization ($ICC_{\text{class}} = .033$) and ROArole of the historical agent ($ICC_{\text{class}} = .104$). These ICCs justify multilevel modelling for substantive analyses. The school-level variance is consistent with between-school differences in history instruction, curricular emphasis, and student composition.—~~The corresponding design effects for the primary composite were modest (HPT_CTX6: $DEFF = 1.56$; HPT_TOT9: $DEFF = 2.16$; see Section 3.4 for computation and implications), although the present design cannot identify which of these mechanisms produced the variation.~~

4.6 DIF Analysis

Table 5 presents omnibus DIF results for all nine HPT items. Using a constrained multi-group graded response model with likelihood-ratio tests and Bonferroni-adjusted significance ($\alpha = .01$), no items were flagged~~for either uniform or non-uniform DIF~~. All adjusted p -values exceeded .01,~~indicating statistically equivalent item parameters across ideology groups. This means that at equal levels of underlying HPT competence, students with low and high ideological attitudes responded to each item in the same way—no~~

individual item functioned differentially as a function of political attitudes. No DIF was detected between the low- and high-ideology groups with this sample and procedure; the magnitude that the design could reliably detect is unknown.

Table 5. DIF Results per HPT Item (Likelihood-Ratio Tests from Multi-Group Graded Response Model)

Item	χ^2	df	p (raw)	p (Bonferroni)	Flagged
POP1	10.46	4	.033	.300	No
POP2	9.01	4	.061	.547	No
POP3	9.72	4	.045	.408	No
ROA1	9.59	4	.048	.432	No
ROA2	3.13	4	.536	1.000	No
ROA3	8.25	4	.083	.744	No
CONT1	1.51	4	.825	1.000	No
CONT2	3.60	4	.462	1.000	No
CONT3	9.63	4	.047	.424	No

Note. Likelihood-ratio chi-square statistics from constrained multi-group graded response models. Each test compares a fully constrained baseline (all items equal across groups) to a model freeing discrimination and threshold-category-intercept parameters for the tested item. Low ($n = 96$) and High ($n = 9695$) ideology groups formed by tertile split on composite ideology score (middle tertile excluded). Bonferroni-adjusted $\alpha = .01$ (9 tests). 0 of 9 items flagged. POP1 showed the largest DIF statistic ($\chi^2 = 11.1810.46$, $p_{\text{raw}} = .025.033$),

well above the Bonferroni threshold after correction ($p_{\text{adj}} = .222.300$). GRM item discrimination and ~~threshold~~category-intercept parameters are reported in Table S1 in the OSF supplementary materials.

~~Although no individual item reached significance after Bonferroni correction, five of nine items (POP1, POP3, ROA1, ROA3, CONT3) had raw p -values below .10. This clustering of near-significant results is consistent with a pattern in which multiple items share weak DIF in the same direction—precisely the scenario under which the all-others-as-anchor strategy without iterative purification is most vulnerable to anchor contamination. Because the anchor set may itself be contaminated by small uniform DIF, the likelihood-ratio tests may underestimate the true magnitude of item-level differences. This pattern does not invalidate the overall null DIF finding, but it suggests that the absence of flagged items should~~undetected DIF among those anchors could affect each test. The result should therefore be interpreted as “no ~~moderate-to-large DIF detected~~” rather than DIF detected with this sample and procedure, not “no DIF present.”

4.7 Measurement Invariance

Table 6 presents the stepwise MG-CFA results. The configural model ~~demonstrated adequate fit (produced~~ CFI = ~~.978~~.977, RMSEA = ~~.043~~.044, and SRMR = .074), ~~confirming that the three-factor structure holds separately in both ideology groups. Constraining factor loadings to equality across groups (metric invariance) produced no meaningful deterioration (.~~ Adding equality constraints did not worsen global fit:

metric $\Delta\text{CFI} = .000, -.001$ and $\Delta\text{RMSEA} = -.002$). Constraining both loadings and thresholds (scalar invariance) similarly maintained fit ($.001$; scalar $\Delta\text{CFI} = +.002, .005$ and $\Delta\text{RMSEA} = -.006$). Both transitions satisfy the established criteria of $|\Delta\text{CFI}| \leq .01$ and $|\Delta\text{RMSEA}| \leq .015$. These changes fall within common fit-index criteria (Chen, 2007; Cheung & Rensvold, 2002), providing evidence for full scalar invariance. All three models converged and passed lavaan's post-estimation admissibility check; the diagonals of their model-implied residual-variance matrices were positive.

Table 6. Multi-Group CFA Fit Indices Across Invariance Levels (WLSMV)

Model	CFI	RMSEA	SRMR	ΔCFI	ΔRMSEA
Configural	.977	.044	.074	—	—
Metric	.976	.043	.081	-.001	-.001
Scalar	.981	.035	.076	+.005	-.008

Note. ΔCFI and ΔRMSEA computed relative to the preceding model. Invariance criteria: $|\Delta\text{CFI}| \leq .01$, $|\Delta\text{RMSEA}| \leq .015$ (Chen, 2007). Low and High ideology groups formed by tertile split on composite ideology ($n \approx 96$ per group); complete item-response data were available for 92 Low and 90 High participants. Items treated as ordered categorical.

The SRMR increased slightly from configural (.074) to metric (.081), marginally exceeding the conventional $\leq .08$ threshold. This threshold was developed for maximum-likelihood estimation with continuous indicators and may not apply directly to WLSMV-estimated models with polychoric residuals; the marginal exceedance should therefore be interpreted cautiously. The SRMR returned to .076 at the scalar step. This pattern can signal

~~partial loading non-invariance. The largest~~ The largest ordinary modification index at
the metric step was ~~11.28 (a suggested cross-loading of 11.22 and suggested a~~ POP2
cross-loading on the ROA factor in the Low ~~ideology group)~~, marginally above the
conventional threshold of 10. This suggests a minor loading difference for POP2 that is
consistent with its weak loading (.30) in the overall CFA (Table 2) rather than evidence of
~~systematic group-specific misspecification~~ group. Because ordinary modification indices
ignore equality constraints, this is a local cross-loading diagnostic, not evidence of a
between-group loading difference.

~~Negative estimated residual variances (Heywood cases) occurred for ROA items~~
~~in the Low ideology group in the configural and metric models. Such estimates~~
~~indicate a non-positive-definite covariance matrix, which can invalidate fit statistics~~
~~and render parameter estimates uninterpretable. While Heywood cases can arise with~~
~~small per-group samples and WLSMV estimation of ordinal items—particularly with~~
~~three-item factors—they signal potential model inadmissibility~~ The admissible solutions
and changes in CFI and RMSEA are compatible with scalar invariance under the fitted
model. This remains qualified evidence rather than a ~~benign estimation artefact.~~ The
per-group sample sizes ($n \approx 82-96$) are at the lower boundary of what MG-CFA with
WLSMV requires. The configural and metric invariance conclusions should therefore be
qualified: they represent the best available evidence given sample size constraints, but
the Heywood cases introduce uncertainty about the precision of the configural baseline.
The invariance results should be treated as convergent evidence alongside the DIF
analysis rather than as definitive confirmation of scalar equivalence. firm equivalence

conclusion: the analysis was added after registration, used only 90–92 complete cases per extreme group, estimated three-indicator factors, and did not model school or classroom clustering.

4.8 Supplementary Invariance: NS-Only Grouping

To verify that invariance findings were not an artefact of the composite ideology grouping variable, we repeated the MG-CFA using groups defined by the NS facet alone (National Socialist sympathy/~~relativisation~~relativization), which represents the most direct theoretical ~~operationalisation~~operationalization of content congruence with the Weimar scenario. With the NS tertile split, group sizes were $n = 126$ (Low) and $n = 125$ (High). ~~The supplementary analysis largely supported scalar invariance, with 37 middle-tertile cases and 7 missing NS scores excluded. Fit did not deteriorate when equality constraints were added~~ (configural CFI = ~~.960~~.959, RMSEA = ~~.050~~.051, SRMR = ~~.070~~.071; metric $\Delta\text{CFI} = +.002$, $\Delta\text{SRMR} = +.006$; scalar $\Delta\text{CFI} = +.014$, $\Delta\text{SRMR} = -.003$). ~~The scalar ΔCFI of .014 marginally exceeds the conventional $\leq .01$ threshold, indicating that full scalar invariance is not unambiguously confirmed for the NS-only grouping. The~~ Because CFI increased and RMSEA and SRMR decreased at the scalar step, the positive ΔRMSEA remained within acceptable bounds, and the direction of CFI change was positive (improved fit), which is atypical for genuine invariance violations. Nevertheless, given that the NS facet is the most theoretically proximal grouping variable and the larger per-group sample sizes ($n = 126/125$) make the criterion breach more interpretable, partial scalar invariance—with possible threshold non-equivalence for individual items—cannot be ruled out. CFI does not indicate worse fit and should not be

716 compared with a deterioration threshold by absolute magnitude. This post-registration
717 analysis provides descriptive robustness evidence, not a second confirmatory test. Full fit
718 statistics for this supplementary analysis are available in the OSF repository.

719 **4.9 Convergent Validity**

720 Scenario-relevant historical knowledge correlated positively with all HPT outcomes: r
721 ~~= .34 (HPT_CTX6)~~for the six-item contextualization composite, $r = .22$ ~~(CONT)~~,for
722 contextualization, and $r = .32$ ~~(POP_rev)~~for reversed present-oriented perspective.

723 These moderate correlations indicate that students with stronger knowledge of the
724 relevant historical period scored higher on HPT, ~~while also demonstrating that HPT is~~
725 ~~not reducible to knowledge alone—the shared variance (approximately 5–12%) leaves~~
726 ~~substantial HPT variance attributable to reasoning processes beyond factual recall. The~~
727 scores were not empirically interchangeable, but their unshared variance cannot be
728 attributed specifically to reasoning because it also includes measurement error and other
729 unmeasured influences. A limitation of this evidence is that the KN items were written

730 to cover content directly relevant to the HPT scenario, so the correlation may partly
731 reflect shared specific-content activation rather than the broader “knowledge-dependent
732 competence” relationship posited theoretically. A general historical knowledge measure
733 would provide a stronger convergent validity test.

734 History grade also correlated with ~~HPT_CTX6~~the six-item contextualization composite
735 ($r = -.153$, $p = .010$; negative sign reflects the Czech scale where 1 = excellent). By con-
736 trast, ~~ideology measures showed negligible correlations with HPT (FR-LF-HPT_CTX6:~~

right-authoritarian attitudes showed a negligible correlation with the composite ($r = -.05$), and social desirability was unrelated to HPT outcomes ($r = -.02$ to $.01$). ~~This pattern—significant knowledge correlations, null ideology and social desirability correlations—is consistent with an instrument that measures disciplinary reasoning rather than attitudes or impression management.~~ One exploratory exception was a negative association between authoritarianism and reversed present-oriented perspective ($r = -.13$, 95% CI $[-.25, -.02]$). The corresponding mutually adjusted standardized KSA-3 coefficient was $\beta = -.166$, $SE = .066$, unadjusted $p = .013$. This association was not preregistered or subjected to the preregistered multiplicity procedure and should be treated as a hypothesis-generating pattern consistent with the reduced-integrative-complexity pathway rather than a confirmed effect. Overall, the positive knowledge associations and absence of detected associations for the primary ideology–HPT relationship are consistent with, but do not establish, the intended score interpretation.

Table 7. Zero-Order Pearson Correlations Among Key Variables

	HPT	Context	Presentism (reversed)	Knowledge	Right-authoritarian attitudes	Authoritarianism	Social desirability
HPT composite	—	[.78, .86]	[.71, .81]	[.23, .43]	[−.16, .07]	[−.15, .09]	[−.14, .10]
Contextualization	.82***	—	[.14, .36]	[.11, .33]	[−.10, .13]	[−.05, .19]	[−.11, .13]
Present-oriented perspective (rev.)	.76***	.26***	—	[.20, .41]	[−.21, .02]	[−.25, −.02]	[−.16, .07]
Knowledge	.34***	.23***	.31***	—	[−.22, .01]	[−.07, .16]	[−.09, .15]
Right-authoritarian attitudes	−.05	.02	−.10	−.10	—	[.43, .60]	[−.30, −.08]
Authoritarianism	−.03	.07	−.13*	.05	.52***	—	[−.26, −.03]
Social desirability	−.02	.01	−.05	.03	−.19**	−.15*	—

Note. Correlations are below the diagonal and unadjusted descriptive 95% confidence intervals are above it. Pairwise complete observations ($n = 276-287$). HPT POP-282-287; exact values in Table S3b). Present-oriented perspective is reverse-scored. KN Knowledge = historical knowledge (0-6 sum); FR-LF = right-authoritarian ideology (1-5); KSA-3 = authoritarianism (1-5); SDR-5 = social desirability (1-5). Intervals and p -values do not adjust for school or classroom clustering. *** $p < .001$. ** $p < .01$. * $p < .05$.

Figure 1 visualizes the near-zero ideology association and positive knowledge association with the six-item contextualization composite.

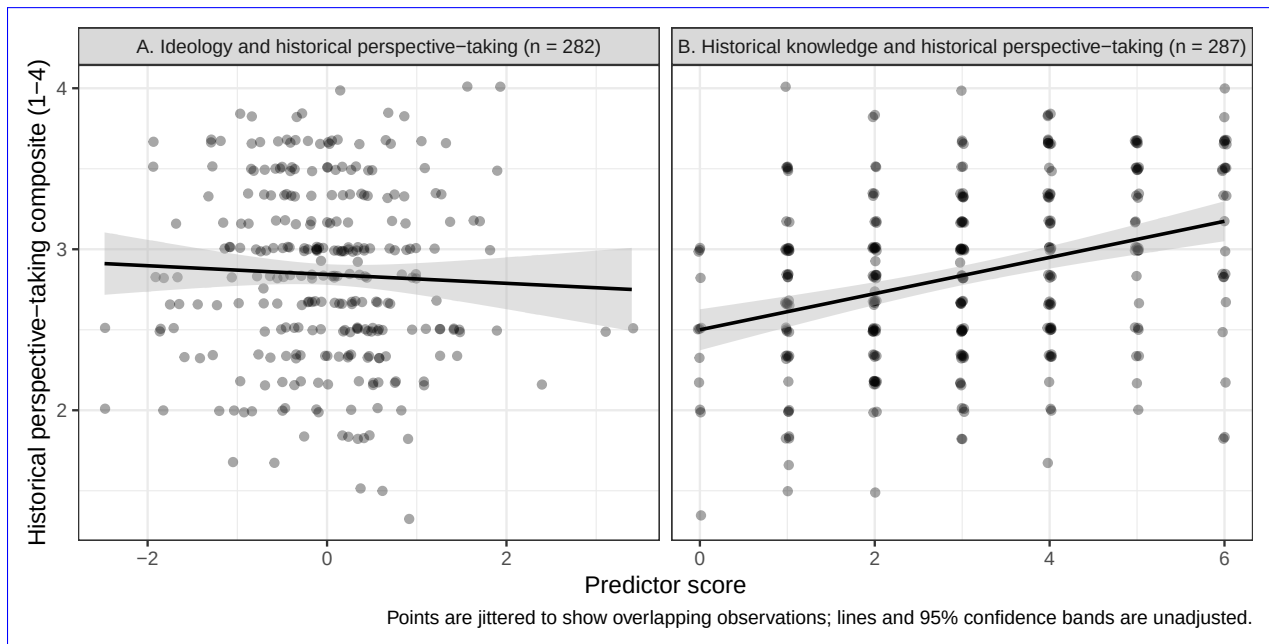


Figure 1: Observed relationships of the six-item contextualization composite with the combined ideology score—the mean of standardized right-authoritarian attitude and authoritarianism scores—(Panel A) and historical knowledge (Panel B). Points are jittered to display overlap; lines and shaded regions show unadjusted linear fits and 95% confidence intervals.

4.10 Substantive Context: Ideology-HPT Relationships

~~Ideology was unrelated to HPT scores across all model specifications: FR-LF~~ The

mutually adjusted standardized right-authoritarian attitude coefficient was near zero for the primary six-item contextualization composite: $\beta = 0.012$, $SE = 0.067$, 95% CI $[-0.120, 0.143]$, $p = .863$ for the primary outcome (HPT_CTX6). (analysis $n = 282$). The fitted model issued a singularity warning because the classroom random-intercept variance was estimated as zero; the school intercept and fixed effects remained estimable, and school fixed-effects sensitivity models gave the same substantive conclusion. Historical knowledge was the only consistent predictor ($\beta = 0.21$ – 0.34 , $p < .001$). As an exploratory sensitivity analysis, equivalence tests used the Two One-Sided Tests (TOST) procedure, which reverses the logic of null-hypothesis testing by asking whether an effect is small enough to be declared practically negligible. The Smallest Effect Size Of Interest (SESOI) was set post hoc at ± 0.20 standardised-standardized units, corresponding to Cohen's (1988) threshold for a "small" effect. This SESOI is wider than the preregistered power target of $\beta \geq .15$, meaning that effects between .15 and .20 cannot be confidently excluded; the TOST results should therefore be treated as exploratory rather than confirmatory. With the ± 0.20 boundary, the TOST results indicated that ideology-right-authoritarian attitude effects at or above $\beta = .20$ are statistically ruled out (all for the primary relationship (TOST p s $< .003 = .0034$; 90% CI for the primary outcome $[-0.10, 0.12]$, entirely within the ± 0.20 SESOI). The ideology-coefficient-right-authoritarian attitude coefficient for the primary composite remained non-significant across all sensitivity checks, including alternative HPT scoring composites, different ideology operationalizations, exclusion of high social desirability responders, random-slope models, and school fixed-effects specifications (all $ps > .43$). These substantive findings complement the measurement evidence: the instrument not

~~only measures equivalently across ideology groups (DIF, invariance) but also yields~~
~~scores that are uncontaminated by political attitudes at the composite level~~no DIF or
composite-level ideology association was detected within the observed ideological range.

5. Discussion

5.1 Summary of Findings

The ~~HPT instrument passed every fairness test applied. Factor structure replicated,~~
~~DIF was absent, scalar invariance held, and ideology~~ results provide qualified fairness
evidence. The expected factor structure was supported, no DIF was detected with this
procedure, and right-authoritarian attitudes did not predict ~~HPT scores. This is the first~~
~~psychometric evaluation of a Czech adaptation of the Hartmann and Hasselhorn HPT~~
~~instrument and the first investigation of ideological fairness for any HPT measure~~the
primary HPT composite; fit-index changes were compatible with scalar invariance,
subject to small extreme groups and unmodelled clustering. The Czech adaptation
~~demonstrated the theorized three-factor structure, acceptable reliability for three-item~~
~~subscales, meaningful convergent validity with~~ showed modest subscale reliability, a
positive relationship with scenario-relevant historical knowledge, and sensitivity to
school-level variation. ~~Historical knowledge was the only construct that predicted~~
~~HPT scores; political ideology, authoritarianism, and social desirability were all~~
~~unrelated~~An unadjusted exploratory authoritarianism-reversed-presentism association
was not subjected to the preregistered multiplicity procedure; other ideology and
social-desirability associations were not detected, although weak reliability constrains the

latter null associations.

5.2 Factor-Structure Evidence

The correlated three-factor model (POP, ROA, CONT) fit ~~substantially better than~~
~~alternatives, confirming that the Czech adaptation preserves the dimensional structure~~
~~of the Czech data better than the tested alternatives.~~ This does not demonstrate
cross-cultural equivalence, and the specification differs from the two-dimensional PCA
pattern in the original German instrument. ~~This replication in a linguistically and~~
~~educationally distinct context confirms that the item grouping holds in a Czech sample~~
~~, though with~~ study. It shows only that the Czech data are compatible with separating
the three item sets, under a near-saturated model ~~the structural confirmation is relatively~~
~~undemanding (see Section 4.1)~~with limited opportunity to reveal misfit. CONT items
loaded most consistently (.61–.72), while POP2 and ROA2 showed weaker loadings
(.30); these were retained for comparability but are candidates for revision in future
adaptations.

5.3 Addressing Modest Reliability

The subscale reliabilities ($\alpha = .47.46$ –.64; $\omega = .55.53$ –.69.70) fall below conventional
benchmarks ~~but are structurally constrained by the three-item design—a feature of~~
~~the original instrument, not a deficiency of the Czech adaptation.~~ With ~~k~~ and limit
individual-level interpretation. Three-item scales make high coefficients difficult to attain,
but brevity does not make the observed precision adequate. Loewinger (1957) cautioned
that highly homogeneous item sets can reduce a construct to repeated versions of a

narrow question and argued for broad content representation; that principle supports balancing coverage and homogeneity, not treating low reliability as harmless. The six- and nine-item HPT scores were more precise ($\omega = .66$ and inter-item correlations of $.22-.37$), Spearman-Brown projections yield $\alpha = .64-.79$ for six-item subscales. Despite modest reliability, the instrument detected theoretically expected knowledge-HPT relationships ($r = .22-.34$) and discriminated between HPT levels in the expected ordering. The disattenuated ideology-HPT correlation ($r = -.065$) confirms that even with perfect reliability, no meaningful ideology association would emerge. We recommend using composite scores (HPT_CTX6 or HPT_TOT9) so we recommend composite scores rather than individual subscales for applied purposes. group-level research. Associations involving the least reliable measures, especially SDR-5, should be read cautiously because attenuation may obscure weak relationships.

5.4 Measurement Fairness Evidence

Scalar invariance was confirmed for the composite ideology grouping and largely supported for the NS-only analysis (with the caveat that the NS scalar ΔCFI marginally exceeded the conventional threshold; see Section 4.8). These results indicate that the HPT instrument imposes largely equivalent measurement demands on students regardless of their political attitudes. Fit index changes were compatible with scalar invariance for both ideology groupings, and the composite-group models were admissible in the current reproducible analysis. The small extreme groups, unmodelled clustering, and post-registration status nevertheless prevent a firm equivalence conclusion. The DIF analysis complements this at the item level: no item exhibited differential functioning

~~after Bonferroni correction, which is particularly important for the CONT items~~
~~most theoretically susceptible to ideological contamination, though the clustering of~~
~~near-significant raw p -values warrants caution (see Section 4.6).~~ adds item-level evidence,
but no design-specific power analysis was conducted and the unidimensional GRM
does not match the CFA structure. Together, these results provide qualified rather than
definitive fairness evidence.

The correlation pattern—positive scenario-relevant knowledge–HPT associations ~~, null~~
~~ideology correlations,~~ (Chapman, 2021; Hartmann & Hasselhorn, 2008; Wineburg, 2001),
a near-zero primary ideology correlation, and null social desirability correlations—is con-
sistent with the intended score interpretation (American Educational Research Associa-
tion et al., 2014; Messick, 1995). However, the convergent evidence is bounded by the
scenario-proximal nature of the knowledge measure (see Section 4.9), which limits the
strength of the knowledge-dependent competence claim.

The near-zero ideology–HPT correlation ($r = -.05$) should not be interpreted as conclu-
sive evidence that ideology plays no role in HPT performance. As discussed in Section
2.3, two opposing pathways are theoretically plausible: a congruence pathway (whereby
attitudinal alignment inflates scores) and a reduced integrative complexity pathway
(whereby authoritarian cognition depresses ~~contextualisation~~contextualization). If both
pathways operate simultaneously, their effects could cancel, producing a near-zero net
correlation that masks two moderate-sized opposing effects. The present data cannot
distinguish a genuine null from cancelling pathways.

The present fairness evidence applies to mainstream Czech adolescent samples with re-

871 stricted ideological range. Users in contexts with greater political ~~polarisation~~polarization
872 should conduct their own DIF and invariance analyses before assuming equivalent mea-
873 surement.

874 5.5 Implications for Assessment Practice

875 The core practical message is that politically charged assessment content ~~does not~~
876 ~~automatically introduce~~need not produce detectable construct-irrelevant ~~variance—but~~
877 ~~this conclusion~~variance in every application. In this study, no primary composite
878 association or DIF flag was detected at the available sensitivity. This pattern is bounded
879 by the ~~present sample's~~ restricted ideological range and should not be generalized beyond
880 the single structured, closed-ended ~~formats~~scenario without further evidence.

881 The evaluation protocol used here ~~provides an evidence-based model~~illustrates a possible
882 workflow for evaluating fairness when assessment content intersects with respondents'
883 attitudes. ~~This protocol is not specific to HPT or history education; it applies to any~~
884 Although developed for HPT, the workflow may inform studies of other scenario-based
885 assessments where content carries attitudinal ~~valence—moral reasoning instruments~~
886 ~~using politically divisive dilemmas, science literacy assessments touching evolution or~~
887 ~~climate change, and civic education assessments involving controversial policies~~valence.

888 The general approach involves five steps:

- 889 1. Identify the attitudinal dimension theoretically most likely to produce CIV.
- 890 2. Form extreme groups on that dimension.
- 891 3. Test DIF at the item level.

892 4. Test measurement invariance at the model level.

893 5. Examine ~~convergent and discriminant correlations to verify that construct scores~~
894 ~~relate to theoretically expected variables and not to the attitudinal dimensiona~~
895 prespecified set of expected relationships, including whether HPT dimensions
896 relate differentially to external outcomes as theory predicts.

897 The format itself may be a fairness mechanism. If scenario-based formats channel delibera-
898 tive processing and override automatic attitude-retrieval (Evans & Stanovich, 2013), then
899 the instrument format—not just the construct—buffers against motivated reasoning. How-
900 ever, this hypothesis has an important implication: if the format itself suppresses attitu-
901 dinal responding, then the present fairness conclusion is conditional on the closed-ended
902 Likert format and cannot be extended to open-ended HPT assessments where ideology
903 could operate through different cognitive pathways.

904 Several design features may explain the instrument’s resistance to attitudinal contamina-
905 tion: the scenario provides contextual information that channels deliberative processing,
906 the Likert items require evaluating propositions about a historical figure rather than ex-
907 pressing personal attitudes, and the school assessment context may activate an academic
908 frame. These are hypotheses, not conclusions—the present study was not designed to test
909 them experimentally, and doing so would require comparing structured and open-ended
910 formats using the same content.

911 An alternative explanation for the ~~observed measurement equivalence compatible~~
912 invariance fit pattern deserves consideration. In a school context where the scenario
913 involves a Nazi-adjacent historical figure, students may uniformly avoid endorsing

presentist responses—not because they are engaging in genuine historical reasoning, but because ~~contextualisation~~ contextualization is the socially safe response regardless of one’s actual political attitudes. If task-demand compliance produces equivalent response patterns across ideology groups, the resulting ~~measurement~~ apparent invariance would reflect shared social desirability pressure rather than genuine absence of ideological influence on reasoning. The null SDR-5 correlations provide partial evidence against this interpretation, but a brief self-report measure of social desirability may not capture the specific task demands operating in a classroom assessment of politically sensitive historical content.

5.6 Limitations

Several limitations qualify these findings.

1. *Single country and scenario.* ~~The study uses one national context and one historical scenario ; whether the fairness findings generalise to other contexts remains to be established.~~ Nine ratings of one Weimar scenario do not provide nine independent samples of the HPT content domain, so scenario-specific wording, knowledge, or moral salience cannot be separated from HPT. Future studies should use multiple scenarios spanning different periods, places, and political contexts.
2. *Restricted ideological range.* ~~Only 17.7%~~ Depending on the measure, 18.0% (FR-LF), 37.5% (KSA-3), or 25.1% (their raw-score mean) of respondents scored above the scale midpoint ~~on ideology measures~~, so the DIF and invariance analyses compare moderately elevated to low ideology, not genuine political extremism. The null ide-

ology finding should be interpreted as “no evidence of CIV in this range-restricted sample” rather than as a general demonstration that politically charged content is fair.

3. *Limited DIF power.* With $n \approx 96$ per group, ~~the analysis was powered primarily to detect moderate-to-large effects ($\Delta\beta \geq 0.5$ logits); weak DIF effects cannot be confidently excluded~~DIF power was limited. A formal power simulation for the GRM likelihood-ratio test at Bonferroni-corrected α was not conducted; ~~future work should quantify, so~~ the minimum detectable DIF magnitude ~~for this design is~~unknown.

4. *Gender imbalance.* The sample was 65.5% male. ~~Gender is a documented moderator of authoritarian attitude endorsement in adolescent samples,~~ and no DIF or sensitivity analysis by gender was conducted. ~~If male students drive both ideology variance and the null finding, the fairness conclusion may not hold for~~The fairness conclusion should therefore be replicated in a more gender-balanced samples.

5. Convenience sampling and unavailable socioeconomic data. The sample over-represented upper-secondary students and was not probabilistic. Because no defensible socioeconomic indicator was collected, neither socioeconomic coverage nor selection along that dimension can be assessed.

6. *Modest reliability.* Subscale reliabilities ($\alpha = .47-.64$) constrain individual-level score interpretations; future adaptations should consider expanding subscale length. Omega hierarchical (ω_h) from the bifactor model was not computed; this index would clarify what proportion of subscale variance is attributable to the general

HPT factor versus specific subscale factors.

7. *Method and format constraints.* All measures were self-report instruments administered in a single session. The closed-ended Likert format may not generalise to open-ended HPT assessments where ideology could operate through different cognitive mechanisms (Smith, 2017; Smith, Breakstone, & Wineburg, 2018).

8. *No response process evidence.* ~~The critical unanswered question is whether Czech students actually invoke historical reasoning when responding or whether the structured format produces contextualising responses through a different cognitive pathway. Cognitive interview studies could resolve this.~~ Think-aloud or cognitive- interview studies, extending Huijgen et al. (2017), could compare the same content in rating and open-response formats. Evidence use and integration of conflicting historical considerations would support contextual reconstruction; explicit moral evaluation or affective engagement would identify processes outside this instrument's narrow cognitive scoring target; paraphrase and option- matching comments would indicate format-driven responding; and endorsement tied to one's present political position would indicate attitude congruence. These observable indicators could distinguish pathways that the present ratings cannot separate.

5.7 Recommendations

For researchers developing or adapting HPT instruments: (a) report ω alongside α for short subscales, with Spearman-Brown projections for context; (b) evaluate fairness through DIF and measurement invariance for theoretically relevant grouping variables;

(c) consider revising weak-loading items (POP2, ROA2) while preserving the theoretical structure; (d) expand subscale length to improve reliability; and (e) replicate fairness evaluations across diverse samples, scenarios, and cultural contexts.

5.8 Conclusion

The Czech ~~adaptation data were compatible with a correlated three-factor representation~~ of the Hartmann and Hasselhorn HPT ~~instrument preserves the three-factor structure of the original, produces item sets and produced~~ scores that relate to scenario-relevant historical knowledge in theoretically expected ways, ~~and functions with largely equivalent measurement properties across students with different political attitudes.~~ Fit-index patterns were compatible with measurement invariance across ideology groups, but small extreme groups, unmodelled clustering, and post-registration analysis prevent a firm equivalence conclusion. At the sensitivity levels afforded by the present sample and in the observed ideological range, ~~political ideology was unrelated to HPT scores across all model specifications.~~ right-authoritarian attitudes were unrelated to the primary HPT composite. An unadjusted exploratory authoritarianism association with one subscale was not subjected to the preregistered multiplicity procedure.

For researchers and assessment developers, the practical implication is that structured scenario-based assessments can ~~employ politically sensitive historical content without detectable measurement bias—at least in mainstream classroom settings with restricted ideological range~~ be evaluated for attitude-related measurement differences using complementary item-, model-, and relationship-level checks. In this sample, those

~~checks detected no bias in the primary composite at the available sensitivity.~~ Whether
this ~~conclusion-pattern~~ extends to more ~~polarised~~-polarized populations, open-ended
assessment formats, or different historical scenarios remains an open empirical ques-
tion. The evaluation protocol ~~demonstrated~~-illustrated here—CFA, DIF, MG-CFA, and
~~convergent/discriminant correlations—provides a replicable~~-a prespecified network
~~of external relationships—provides a possible~~ template for answering that question
wherever attitudinally charged content meets competence assessment.

Data Availability Statement

De-identified data, annotated analysis scripts, the complete instrument battery (in Czech),
supplementary materials, and all deviations from the original instrument are available on
the Open Science Framework (<https://doi.org/10.17605/OSF.IO/YNG37>). Data are pro-
vided in machine-readable formats (RDS and xlsx) accompanied by a full variable code-
book (PDF). The analysis pipeline consists of seven numbered R Markdown scripts that re-
produce all reported results; a README file documents the directory structure, execution
order, software requirements, and mapping of scripts to manuscript tables and figures.
Supplementary materials (~~Table S1: GRM item parameters~~[Tables S1–S5](#)) are included in
the OSF repository. The study design, hypotheses, and analysis plan were preregistered
on OSF prior to data collection; [deviations and omissions are disclosed in Table S5](#).

Conflict of Interest Disclosure

The author declares that he complies with the PCI rule of having no financial conflicts of interest in relation to the content of the article, and has no non-financial conflicts of interest.

Ethics Statement

This study involved secondary school students completing questionnaires during regular class periods. Under Czech law (Act No. 110/2019 Sb., on the processing of personal data), ethics committee approval is not required for non-interventional educational research involving anonymous questionnaires where no personal data are collected. Participation was voluntary; students were informed about the study's purpose and their right to decline participation without consequences. All responses were de-identified at the point of data entry. No personally identifiable information was collected or stored. The study followed the principles of the Declaration of Helsinki for research involving human participants.

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Author Contributions (CRediT)

Juda Kaleta: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Project administration.

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1222 **Is there a data dictionary and/or readme file included with the data to make it inter-**
1223 **pretable?**

1224 **[X] Yes, available at this link:** <https://doi.org/10.17605/OSF.IO/YNG37> (full variable
1225 codebook in PDF and README file)

1226 ☐ No, justification:

1227 **Do you indicate in the manuscript how the sample size was determined?**

1228 **[X] Yes.** (Section 3.4: a priori power analysis targeting 80% power to detect $\beta \geq .15$)

1229 ☐ No, justification:

1230 **Do you report all data exclusions (e.g., outliers, careless responders)?**

1231 **[X] Yes.** (Section 3.4 and ~~4.6: Table S5: the~~ middle ideology tertile ~~excluded for DIF~~
1232 ~~extreme-group design; all exclusions reported~~~~was excluded for post-registration DIF and~~
1233 ~~MG-CFA analyses; this analysis-specific exclusion is reported.~~)

1234 ☐ No, justification:

1235 **Do you report all inclusion/exclusion criteria and when they were established?**

1236 **[X] Yes.** (~~Inclusion/exclusion criteria~~ Participant eligibility is described in Section 3.2
1237 and was preregistered; post-registration analysis-specific exclusions are described in Sec-
1238 tions ~~3.4 –3.5 and were preregistered prior to data collection~~ and Table S5.)

1239 ☐ No, justification:

1240 **Are all measures, questions, and/or conditions used in the study described in the**
1241 **manuscript or available in the supplemental material?**

1242 **[X] Yes.** (All instruments described in Sections 3.1–3.3; full Czech-language instrument
1243 battery available on OSF: <https://doi.org/10.17605/OSF.IO/YNG37>)

1244 ☐ No, justification:

1245 **Section 2: Analysis Scripts/Code/Codebooks**

1246 **Does your manuscript contain any analysis of quantitative or qualitative data?**

1247 **[X] Yes** (continue with next question)

1248 ☐ No (skip to Section 3)

1249 **Are third-party analysis scripts/code (e.g., R, Stata), codebooks, or other relevant doc-**
1250 **umentation available within a trusted digital repository?**

1251 **[X] Yes, available at this link:** <https://doi.org/10.17605/OSF.IO/YNG37> (7 R Mark-
1252 down pipeline scripts, 4 figure scripts, 2 supplementary analysis scripts, and a Makefile)

1253 ☐ No, justification:

1254 **Are the analysis scripts/code (e.g., R, Stata), codebooks, or other relevant documenta-**
1255 **tion cited in the manuscript, with a DOI?**

1256 **[X] Yes, the DOI is as follows:** <https://doi.org/10.17605/OSF.IO/YNG37>

1257 ☐ No, justification:

1258 **Section 3: Study Materials**

1259 **Does your manuscript contain any research materials (e.g., stimuli, programming code,**
1260 **questionnaires, interview protocols)?**

1261 **[X] Yes** (continue with next question)

1262 ☐ No (skip to Section 4)

1263 **Are all study materials and descriptions of study procedures available within a trusted**
1264 **digital repository?**

1265 **[X] Yes, available at this link:** <https://doi.org/10.17605/OSF.IO/YNG37> (Czech-
1266 language versions of all questionnaires: HPT, FR-LF, KSA-3, SDR-5, and historical
1267 knowledge test)

1268 ☐ No, justification:

1269 **Are all third-party study materials, descriptions of study procedures, or other relevant**
1270 **documents cited in the manuscript, with a DOI?**

1271 ☐ Yes, the DOI is as follows:

1272 **[X] No, justification:** Third-party instruments (~~HPT, FR-LF, KSA-3, SDR-5~~) are cited via
1273 standard bibliography references (Hartmann & Hasselhorn, 2008; ~~Becker~~ Decker et al.,
1274 ~~2020; Lederer, 2002; Stöber, 2001~~2013; Beierlein et al., 2014; Hays et al., 1989); DOIs are
1275 included where available. The adapted Czech versions are deposited on OSF ([https://](https://doi.org/10.17605/OSF.IO/YNG37)
1276 doi.org/10.17605/OSF.IO/YNG37).

1277 **Section 4: Preregistration**

1278 **Were any aspects of your manuscript preregistered?**

1279 **[X] Yes** (continue with next question)

1280 ☐ No (do not complete the rest of the form)

1281 **Does the manuscript contain an accessible link to the preregistration?**

1282 **[X] Yes, available at this link:** <https://osf.io/zsngy>

1283 ☐ No, justification:

1284 **Do you clearly indicate in the manuscript which parts were preregistered and which**
1285 **parts were not?**

1286 **[X] Yes.** (~~Preregistered components include hypotheses H1–H2, primary multilevel~~

1287 ~~models, DIF/MG-CFA analyses, and sample size determination; the TOST equivalence~~
1288 ~~analysis is explicitly labelled “exploratory” in Section 4.10 as it used a post-hoc SESOI~~
1289 ~~wider than the preregistered power target~~Section 2.5 gives a concise disclosure of
1290 post-registration changes and omitted registered analyses. Table S5 distinguishes
1291 registered tests, omissions, implemented modifications, and post-registration analyses.)

1292 ☐ No, justification:

1293 **Are all preregistered analyses reported in the text or linked in the supplemental mate-**
1294 **rial?**

1295 ☐ Yes.

1296 [X] ~~Yes.~~No, justification: ~~(All preregistered hypotheses and analyses are reported in~~
1297 ~~Sections 4.1–4.10 and the supplementary materials)~~ The registered continuous-ideology
1298 MIMIC model, complete moderation and simple-slope analyses, clustered correlation
1299 comparison, complete covariate set, one-tailed tests, Benjamini–Hochberg correction, and
1300 registered missing-data rule were not all implemented. Table S5 reports these omissions
1301 and the analyses used instead.

1302 ~~—☐ No, justification:~~

1303 **Are all deviations from the preregistration plan clearly disclosed in the manuscript**
1304 **(either in text or in a table)?**

1305 [X] **Yes.** ~~(One deviation: TOST SESOI set post hoc at~~ Section 2.5 and Table S5 disclose the
1306 changed inferential hierarchy, omitted registered procedures, instrument discrepancies,

1307 and post-registration additions. TOST and its ± 0.20 instead of the preregistered power
1308 target of $\beta \geq .15$; disclosed in Section 4.10 and the analysis is labelled exploratory SESOI
1309 are explicitly labelled exploratory.)